ZenCro
DIGITAL

ZENCRO SCHOOL MANAGEMENT SYSTEM

Detailed Research, Product Strategy & Technical Blueprint

India-focused K-12 School Management & Operations Platform

PREPARED FOR	ZenCro Digital (internal research and product strategy document)
EDITION	Detailed research edition, expanded from the consolidated 14-page blueprint
DOCUMENT DATE	September 2026
SCOPE	K-12 school management only. College / university ERP is out of scope.
RESEARCH BASIS	Publicly available evidence only. No direct interviews, calls or surveys.

Internal working document. Contains proposals and hypotheses; none of the proposed features are claimed to be implemented.

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How to Read This Document

This document expands the consolidated 14-page ZenCro School Management System blueprint into a detailed research and product strategy reference. It preserves the research conclusions and terminology of that baseline. It does not replace them with generic school-ERP knowledge, and it does not add evidence that the underlying research did not establish.

Two classification systems used throughout

Two separate labelling systems appear in the document. The first describes **how strong a statement is** (evidence labels). The second describes **where a product capability sits in the plan** (feature-status labels). They are never interchangeable: a feature can be an INITIAL MVP item and still rest on a hypothesis rather than on verified evidence.

Evidence labels (how strong is this statement?) run from VERIFIED FACT and REGULATORY / OFFICIAL, through VENDOR CLAIM, customer, independent and institutional evidence and ANALYST OBSERVATION, to ZENCRO HYPOTHESIS / RECOMMENDATION and UNKNOWN. Definitions and handling rules are in Section 3.3. UNKNOWN is never silently converted into a fact and never treated as NO.

Feature-status labels (where does this capability sit in the plan?)

Status	Meaning	Sections
INITIAL MVP / STAGE 1	Proposed for the first build. A proposal, not a claim of existing functionality.	14, 15
PLANNED LATER (V1)	Intended after the core MVP is stable. Planned, not implemented.	17
FUTURE / VALIDATION-DEPENDENT	Possibly valuable, but current public evidence does not justify building it yet.	18
NOT PART OF INITIAL MVP / STAGE 1	Scope-control decisions. These items must not expand the first build unless a new requirement is explicitly approved.	16

Scope statement
This document covers K-12 school management only. College and university ERP are out of scope.

1. Executive Summary

The Indian K-12 school ERP market is crowded. Across the vendors reviewed, the core capabilities of a school management system (student information, fees and payments, attendance, examinations and report cards, parent and teacher access, communication and reporting) are already widely offered. The research therefore does **not** establish that ZenCro has found an unserved ERP category, and this document does not claim that ZenCro has no competitors.

The stronger conclusion is that any opportunity is more likely to lie in **execution quality**: making common school workflows simpler, more reliable, more portable and better adapted to how Indian schools actually operate. Publicly available customer and institutional evidence points repeatedly to friction in areas where software already exists on paper: schools continuing to run hybrid combinations of ERP, spreadsheets, WhatsApp and paper; communication that is hard to target; mobile experiences that frustrate users; support that customers describe as slow; reconciliation of fee collections; and the effort of migrating between systems.

The principle that follows is that **more features do not make a better product**. The proposed ZenCro first stage is deliberately small: one coherent operating loop for a school rather than a broad module catalogue.

Key findings

#	Finding	Status
1	Core K-12 ERP capabilities are broadly commoditized across Indian vendors. Feature parity alone is not a differentiator.	[ANALYST OBSERVATION]
2	UDISE+ 2024-25 gives a strong baseline for the scale of the school system, but not for ZenCro's target segment, willingness to pay or product-market fit.	[REGULATORY / OFFICIAL]
3	Recurring public evidence points to workflow friction (hybrid tooling, communication, mobile usability, support, connectivity, PTM, paper forms, reconciliation, migration). These are recurring themes, not universal conditions.	[CUSTOMER EVIDENCE] [ANALYST OBSERVATION]
4	Potential execution opportunities exist in fee reconciliation, targeted communication, teacher workflow simplicity, migration and export, digital forms and consent, PTM booking, integrations and selected offline workflows. They are potential opportunities, not guaranteed gaps.	[ANALYST OBSERVATION] [ZENCRO HYPOTHESIS]
5	International K-12 products are a source of workflow ideas (teacher and parent UX, unified communication, digital forms, PTM booking, migration pipelines, student 360 views). They must be adapted to India, not copied.	[ANALYST OBSERVATION]
6	Willingness to pay, ROI, adoption rates, implementation duration and SLA expectations are unknown.	[UNKNOWN]

Recommended first stage

The recommended first stage is a focused operational core, labelled throughout this document as **INITIAL MVP / STAGE 1**: school setup, student management, attendance (with selected offline capture), fees, exams and results, communication, a parent dashboard, PTM scheduling, basic digital forms, import and export, and role-specific dashboards. Payment gateways, admissions, homework, advanced timetabling and all AI capabilities are deferred.

The technical direction mirrors that discipline: a modular monolith built with Next.js, React and TypeScript, with Supabase providing PostgreSQL, Auth, Row Level Security, Storage and selective Realtime, deployed on Vercel. Offline support is limited to teacher attendance and gated by a proof of concept. Separate backends, microservices, Kubernetes, Redis, message queues and vector databases are explicitly excluded from the first stage.

> **What this document does not claim**
> It does not claim a market without competitors, a validated target customer, a proven willingness to pay, any ROI or time-saving figure, any competitor pricing that was not publicly disclosed, any competitor technology stack, DPDP compliance, or that any proposed feature already exists.

Decisions requested from ZenCro leadership

- Confirm the Stage 1 scope boundary and the exclusion list in Section 16 as the working scope-control baseline.
- Confirm that direct validation with prospective schools (interviews, workflow walk-throughs, pricing conversations) is the next step before commercial commitments. See Section 28.4.
- Approve the attendance offline proof of concept (Section 21) as a gated engineering experiment.
- Assign ownership for the DPDP legal/product workstream (Section 23.2) before any commercial deployment.

2. Research Objectives

The research was organised around six objectives. Each objective was framed to separate what is established (market structure, common capabilities) from what is only plausible (execution gaps, target segment) and from what is unknown (commercial parameters).

Objective	Guiding questions	Addressed in
Understand the structure of the Indian K-12 market	How large is the school system? Which management types exist? What does the official baseline establish and what does it leave open?	Section 4
Map common capabilities and positioning across Indian vendors	Which capabilities are table stakes? Which claims (offline, AI, multi-campus) are selective? How should pricing be treated when it is not public?	Sections 5, 6
Benchmark international K-12 practices for workflow ideas	Which practices reduce administrative friction and might be adapted, and what India-specific adaptation is needed?	Section 7
Identify recurring problems from public customer evidence	Which operational themes recur across reviews, discussions, institutional and procurement material? How strong is each?	Sections 8, 9
Separate established features from potential execution gaps and unvalidated opportunities	Where is the market commoditized? Where might execution differ? Where is evidence too weak to act?	Section 10
Translate findings into scope, architecture and roadmap	What should be built first, what later, what only after validation, and on which technical foundation?	Sections 11-27

Research boundaries

- **In scope:** K-12 school management and operations in India; public-evidence research; product and technical direction for a first stage.
- **Out of scope:** college and university ERP; primary research with schools; pricing derived from assumption; quantified ROI models.
- **Interpretation rule:** where the research supports a theme but not its magnitude, the theme is reported without inventing a magnitude.

3. Research Methodology & Evidence Standards

3.1 Approach

The research relied exclusively on publicly available evidence. No direct interviews, calls or surveys were conducted with schools, administrators, teachers or parents. This is a defining limitation: public-web evidence is not statistically representative, and vocal customers, whether satisfied or dissatisfied, are over-represented compared with the silent majority of users.

Evidence was gathered across twelve source categories and then synthesised by theme rather than by individual source, so that a conclusion is only recorded as a recurring theme when it appeared across more than an isolated mention. Each statement in this document is then labelled with the evidence class that best describes it.

3.2 Source categories and how each was used

Source category	Used for	Typical label	Main limitation
Government and regulatory sources; Ministry of Education; UDISE+	Market baseline and scale	[REGULATORY / OFFICIAL]	Describes the school system, not buyers or willingness to pay.
Official vendor product pages	Capability patterns and positioning	[VENDOR CLAIM]	Marketing language; not independently verified.
Vendor technical documentation	How capabilities are described technically	[VENDOR CLAIM]	Documentation may lag or lead the actual product.
School and institutional material	Workflows and requirements in real schools	[INSTITUTIONAL EVIDENCE]	Individual institutions are not representative of all schools.
Procurement / RFP-style documents	Requirement patterns, integration and reporting needs	[INSTITUTIONAL EVIDENCE]	Skewed toward larger or chain institutions.
Independent software-review platforms	Customer experience themes	[INDEPENDENT EVIDENCE]	Self-selected reviewers; possible incentivised reviews.
App-store evidence	Mobile usability and reliability themes	[CUSTOMER EVIDENCE]	Reflects app users; ratings shift over time.
Public customer discussions	Problem language and workarounds	[CUSTOMER EVIDENCE]	Used cautiously; anecdotal.
Case studies	Reported implementation outcomes	[VENDOR CLAIM]	Usually vendor-authored; treated as vendor claims.
Technical documentation (open-source and other)	Architecture and implementation patterns	[ANALYST OBSERVATION]	Patterns to learn from, not competitor stack inference.
International school-management products	Workflow benchmarking	[ANALYST OBSERVATION]	Different market conditions; adapt, do not copy.

3.3 Evidence labels and handling

Label	Definition	Handling rule in this document
[VERIFIED FACT]	Directly supported by a reliable source.	May be stated plainly, with the source category identified.
[REGULATORY / OFFICIAL]	Government, regulator or official institutional evidence.	Used for the UDISE+ baseline. Interpreted narrowly.
[VENDOR CLAIM]	Stated by a product or vendor, not independently validated.	Always attributed to the vendor. Never restated as independently verified.
[CUSTOMER EVIDENCE]	Customer-reported experience or testimonial.	Used to identify themes. Never used to imply prevalence.
[INDEPENDENT EVIDENCE]	Third-party review or discussion.	Used for problem evidence. Treated as anecdotal at the individual level.
[INSTITUTIONAL EVIDENCE]	School or educational-organisation material.	Used for workflow and requirement evidence.
[ANALYST OBSERVATION]	Synthesis of multiple sources.	Explicitly marked as interpretation.
[ZENCRO HYPOTHESIS] [ZENCRO RECOMMENDATION]	Proposed direction requiring validation.	Never presented as established. Carries an explicit validation note where relevant.
[UNKNOWN]	Insufficient public evidence.	Reported as UNKNOWN. Not converted into a fact. Not treated as NO.

3.4 Working rules applied to the writing

- UNKNOWN information is never converted into a fact.
- Vendor marketing claims are never presented as independently verified facts.
- No statistic is invented. No customer interview, survey result, ROI figure, adoption percentage or competitor technology stack is invented.
- Where competitor pricing is not publicly available, the exact phrase **NOT PUBLICLY DISCLOSED - QUOTE REQUIRED** is used. Competitor pricing is never estimated.
- A competitor technology stack is not inferred from website appearance or technology-detection tools.
- In feature comparisons, YES, NO and UNKNOWN are used carefully. UNKNOWN does not mean NO.
- Future features are not presented as existing. MVP proposals are not presented as built functionality.

3.5 Earlier numeric claims deliberately excluded

Earlier exploratory work produced numeric claims that were later judged not reliable enough to present as market-wide facts. They are listed here so that they are not reintroduced into ZenCro planning material by accident.

Excluded claim type	Why it is excluded	What is reported instead
Exact notification counts	No reliable public basis for a market-wide figure.	Communication overload is reported as a recurring theme.
Hours saved by automation	Would imply an ROI that is UNKNOWN.	Reduced administrative friction is stated as a design goal, not a measured outcome.
Paper loss percentages	Anecdotal; not measurable from public sources.	Paper permissions and forms are reported as a recurring workflow opportunity.
Connectivity duration	Varies by school and location; not established.	Connectivity constraints are reported as supported by public evidence, with prevalence UNKNOWN.
Exact error rates	No reliable public basis.	Fee reconciliation burden is reported as a problem area whose magnitude is not established.

3.6 Limitations

- Not statistically representative; no primary research.
- Vendor material dominates public availability and is inherently promotional.
- Willingness to pay, ROI, retention, implementation duration, SLA performance and adoption rates remain unknown.
- Segment-level preferences (by school size, board, state, fee level, urban or rural setting) are largely UNKNOWN.
- Exact URLs and source-level citations from the earlier research are retained in the project source archive and are not reproduced here (see Section 30).

4. Indian K-12 School Landscape

The Ministry of Education's UDISE+ 2024-25 release provides the most authoritative baseline available for the size and composition of the Indian school system. It is used here as a statement of scale and structure. It is not used to infer a target market.

4.1 UDISE+ 2024-25 baseline

Metric	UDISE+ 2024-25	Share of total schools
Total schools	1,471,473	100%
Students	246,932,680	n/a
Teachers	10,122,420	n/a
Government schools	1,013,322	68.9%
Government-aided schools	79,349	5.4%
Private unaided recognised schools	339,583	23.1%
Other schools	39,219	2.7%
Residential schools	46,517	3.2% (separate metric)

Shares are simple arithmetic on the UDISE+ figures above **[ANALYST OBSERVATION]**. The four management categories (government, government-aided, private unaided recognised, other) sum exactly to the total number of schools. Residential schools are reported as a separate metric and their overlap with the management categories is not stated in the baseline, so they are not added to those categories.

4.2 Derived ratios (arithmetic only)

Ratio	Value	Interpretation
Students per school (national average)	168	A national mean that conceals very large variation between schools. It must not be used to size a ZenCro customer.
Teachers per school (national average)	6.9	Same caution: an average across very different school types.
Students per teacher (national average)	24.4	Indicates scale of teacher workload across the system; not a measure of any single school.

These ratios describe the system as a whole. Schools that would plausibly buy a paid management platform are a subset of this distribution, and the baseline provides no information on which subset.

4.3 What the baseline establishes and what it does not

The figures establish	The figures do NOT establish
The overall scale of the Indian school ecosystem, which is very large. The relative size of government, aided, private unaided and other schools. That private unaided recognised schools form a substantial category (23.1% of schools).	ZenCro's target customer. Willingness to pay for school software. Product-market fit. Exact market share or adoption of any product. Any ROI or savings figure. Which schools already use a management platform.

4.4 Segment interpretation

[ZENCRO HYPOTHESIS] Private unaided recognised schools are a plausible commercial focus to investigate. Their management structure often includes an owner, trust or management committee alongside a principal, which may make purchasing decisions more local than in government systems. This remains a hypothesis. The research does not prove that private unaided schools are the only, or universally best, buyer segment, and it does not quantify how many of them currently use software, which products, or at what price.

[UNKNOWN] National totals also conceal state-level and board-level differences in curriculum, assessment and report-card formats, language, and administrative requirements. The baseline does not quantify these differences. This is one reason the product strategy (Section 11) avoids locking the product to a single board or payment provider.

> **Implication for product planning**
> The UDISE+ baseline supports the statement that the Indian school system is large. It does not support any statement about ZenCro's revenue potential. Commercial sizing requires segment-level evidence that this research did not gather.

5. Indian School ERP Competitive Landscape

The research reviewed a broad set of Indian and India-relevant school-management products. This section records what could be concluded about the landscape as a whole. Product-level detail belongs in the source archive and in the verification worksheet in Appendix A, so that every YES in a comparison can be traced to a source.

5.1 Products reviewed

EduGradUP; Fedena; Entab CampusCare; Vidyalaya ERP; MyLeading Campus; GegoK12; OpenEduCat; Campus 365; Classe365; MyClassboard; Teachmint; SchoolLog / SkoolBeep; ERPMY; and other relevant Indian and India-relevant school-management products.

This list is a research coverage record. It is not a ranking, a market-share statement or an endorsement, and it does not imply that all products serve the same segment.

5.2 Common capability patterns

Capability	Pattern across vendors	Evidence basis
Student information system / school records	Widely offered	[VENDOR CLAIM]
Fees and payment workflows	Widely offered	[VENDOR CLAIM]
Attendance	Widely offered	[VENDOR CLAIM]
Exams, marks and report cards	Widely offered	[VENDOR CLAIM]
Parent and teacher portals / apps	Widely offered	[VENDOR CLAIM]
Notifications and communication	Widely offered	[VENDOR CLAIM]
Role-based access	Widely offered	[VENDOR CLAIM]
Reporting and exports	Widely offered	[VENDOR CLAIM]
Integrations and payment / communication services	Offered, with varying depth	[VENDOR CLAIM]
Multi-campus support	Present in some products	[VENDOR CLAIM]
Offline or low-connectivity capability	Claimed in some products	[VENDOR CLAIM]
AI-assisted timetable, analytics or other automation	Claimed in selected products	[VENDOR CLAIM]

All capability presence in this table is drawn from vendor material, so it is a **vendor claim**. It establishes that a capability is marketed, not that it works well, that it is used widely, or that customers are satisfied with it.

5.3 Reading the pattern

[ANALYST OBSERVATION] The breadth of overlap is the central finding of the competitive review. When a large number of products claim the same feature set, listing more features does not distinguish a product. Vendors then compete on implementation quality, support, usability and price. That is consistent with the recurring customer-evidence themes in Section 8, which concentrate on how things work rather than on what is missing.

Three claim types deserve particular caution because they vary in credibility and are hard to verify from public sources:

- **Offline / low-connectivity claims** [VENDOR CLAIM]: the scope of what works offline, how synchronisation and conflicts are handled, and whether the claim covers attendance, marks or the whole application are generally not verifiable from public material.
- **AI and automation claims** [VENDOR CLAIM]: the degree to which AI features are production capabilities versus marketing positioning is not established.
- **Multi-campus claims** [VENDOR CLAIM]: depth varies, and multi-campus reporting is also visible as a customer and procurement requirement (Section 8).

5.4 Pricing discipline

Public pricing is available for some products and not for others. The rule applied throughout the research and this document is fixed:

> **Pricing rule**
> Where a competitor's pricing is not publicly available, it is recorded as: NOT PUBLICLY DISCLOSED - QUOTE REQUIRED. Competitor pricing is never estimated, averaged or inferred from other products.

Consequences: ZenCro cannot currently claim to be cheaper or more expensive than any competitor. Any price positioning must come from ZenCro's own validated willingness-to-pay work (Section 26), not from competitor extrapolation.

6. Detailed Competitor Capability Analysis

Because the underlying evidence for feature presence is vendor material, the value of a detailed capability analysis lies less in a grid of ticks and more in a framework for judging **execution quality**. This section sets out the rules for reading feature matrices, the domain-by-domain baseline that ZenCro must meet just to be credible, and the execution questions that separate products in practice.

6.1 Rules for feature matrices

Value	Definition	Requirement
YES	A source shows the capability is offered.	A source reference must be recorded. A vendor page is a VENDOR CLAIM, not proof of quality.
NO	A reliable source states the capability is not offered.	Rarely established from public sources; absence of a mention is not evidence.
UNKNOWN	Public evidence is insufficient either way.	Must not be treated as NO. UNKNOWN is the default in the absence of a source.

6.2 Domain baseline and execution questions

For every capability domain the table below states what a buyer is likely to treat as a minimum expectation **[ANALYST OBSERVATION]** and the questions that indicate genuine execution differences **[ZENCRO RECOMMENDATION]**. These questions define how competitor demos, trials and reviews should be assessed, and they translate directly into ZenCro design goals.

Domain	Baseline expectation	Execution questions to test	ZenCro implication
Student records (SIS)	Student profile, class and section, guardian details, status, documents.	How easy is bulk import and clean-up? Can data be exported completely? How are duplicates and status changes handled?	Treat import and export as core features from Stage 1, not add-ons.
Attendance	Daily or class attendance, reports, percentage, parent visibility.	How many taps for a teacher? What happens on a poor connection? How are corrections tracked?	Fast teacher entry, tracked corrections, offline capture proof of concept.
Fees and payments	Fee heads, structure, dues, receipts, outstanding and collection reports.	How clear are balances after partial payments? How are collections reconciled? How are receipts corrected?	Deterministic fee rules, immutable receipts, basic reconciliation in Stage 1.
Exams and report cards	Exams, marks entry, grading, results, report cards.	How flexible are report-card layouts? How long does marks entry take a teacher? How are board formats handled?	Define a bounded, configurable report-card approach; broaden formats later.
Parent and teacher portals	Web or mobile access to child information and class tasks.	How reliable and simple is the mobile experience? How do parents authenticate?	Mobile-friendly web experience first; native apps only after validation.
Communication	Announcements and notifications.	Can messages be targeted precisely? Is there a single channel or many? How is overload avoided?	Audience targeting by class, section and role; external channels later.
Access control, reports, exports	RBAC, standard reports, exports.	Are roles configurable? Are reports customisable? Can all data be exported?	RBAC and RLS from day one; exports for all core entities.
Multi-campus	Present in some products.	How is cross-campus reporting delivered?	Not Stage 1; revisit if target customers require it.
Integrations	Payment and communication services in some products.	Which integrations are prebuilt, and which need custom work?	Integrations are planned later, prioritised by customer demand.
Offline / low connectivity	Claimed in some products.	What exactly works offline? How are conflicts resolved?	Narrow attendance-only POC, with explicit sync status.
AI / automation	Claimed in selected products.	What is a production capability versus positioning?	Deferred; AI never controls core records.

6.3 Pricing and packaging framework

Because competitor pricing is frequently not public, the comparison dimension is the *structure* of pricing rather than the level. When a quote is obtained through a vendor's normal sales channel, the following dimensions should be recorded. Until then every cell that cannot be sourced reads as below.

Dimension	Question	Status without a source
Price level	What does the school pay per year?	NOT PUBLICLY DISCLOSED - QUOTE REQUIRED
Pricing basis	Per school, per student, or tiered?	UNKNOWN unless stated by the vendor
Module bundling	Are core modules bundled, or priced separately?	UNKNOWN unless stated by the vendor
Implementation / migration fee	Is onboarding charged?	UNKNOWN unless stated by the vendor
Integration charges	Are payments or messaging integrations extra?	UNKNOWN unless stated by the vendor
Support tiers	What support level is included?	UNKNOWN unless stated by the vendor

Dimension	Question	Status without a source
Data export terms	Can the school export all data on exit, and at what cost?	UNKNOWN unless stated by the vendor

6.4 Technology-stack policy

Competitor technology stacks were not inferred. Website appearance, page source patterns and technology-detection tools are not treated as evidence of a vendor's architecture, because they are unreliable and can mislead product decisions. Where a vendor publishes technical documentation, what it states is treated as a **[VENDOR CLAIM]** statement about that product. Otherwise the competitor stack is UNKNOWN, and ZenCro's architecture choices in Section 19 are made on ZenCro's own requirements, not on assumptions about competitors.

6.5 Competitive response considerations

[ZENCRO HYPOTHESIS] Any single workflow improvement (for example PTM booking or digital consent) can be replicated by competitors that already have a large installed base. A durable position is therefore unlikely to come from any one feature. The hypothesis is that it may come from a combination that is difficult to bolt onto an existing broad product: a coherent workflow model, consistent quality on the core operating loop, transparent data ownership and migration posture, and a support experience that customers perceive as responsive. Whether this combination is valued enough to influence buying decisions is UNKNOWN and is a primary validation target (Section 28.4).

7. International K-12 Benchmarking

7.1 Role of benchmarking

International K-12 products were reviewed as a source of workflow ideas, not as a blueprint. Markets differ in payment rails, communication habits, assessment structures, funding and regulation, so a practice that works well elsewhere may not transfer directly to Indian schools.

7.2 Products reviewed

PowerSchool, Infinite Campus, ManageBac, Engage, iSAMS, QuickSchools, Brightwheel and related international products. The research does not attribute individual practices to individual products in this document; practices are presented as a pattern observed across the set **[ANALYST OBSERVATION]**.

7.3 Practices worth evaluating for India

Practice	Why it is relevant	India adaptation required	Placement
Teacher-friendly UX	Teacher time is the scarcest resource in attendance and marks entry.	Low-end devices, poor connectivity, limited training time.	**INITIAL MVP / STAGE 1**
Parent-friendly UX	Parents are a high-volume, low-training user group.	Language and device diversity; WhatsApp habits.	**INITIAL MVP / STAGE 1**
Unified communication	One coherent channel reduces overload and missed messages.	WhatsApp and SMS realities; in-app-only may not reach every parent.	**INITIAL MVP / STAGE 1** (in-app); **PLANNED LATER (V1)** (external channels)
Digital forms	Replaces paper permission slips and acknowledgements.	Who may respond on behalf of a child; retention and legal review.	**INITIAL MVP / STAGE 1** (basic); **PLANNED LATER (V1)** (expanded)
Digital consent workflows	Creates a recorded, retrievable response.	DPDP and school policy implications require legal review.	**INITIAL MVP / STAGE 1** (basic)
PTM booking	Removes manual slot coordination.	Different school PTM formats; teacher availability patterns.	**INITIAL MVP / STAGE 1**
Migration / export pipelines	Lowers switching cost in both directions.	Source data is often Excel or paper in inconsistent formats.	**INITIAL MVP / STAGE 1** (basic); **PLANNED LATER (V1)** (improved tooling)
Prebuilt integrations	Reduces custom work at onboarding.	UPI and payment providers, Tally / Zoho where needed, SMS / WhatsApp providers.	**PLANNED LATER (V1)**
Student 360 view	A single view of a student across modules.	Depends on data quality across attendance, fees and results.	**INITIAL MVP / STAGE 1** (parent view); richer views **PLANNED LATER (V1)**
Activity / audit timeline	Traceability for corrections and changes.	Retention policy and privacy scope.	**INITIAL MVP / STAGE 1** (audit-friendly records)
Teacher-first dashboards	Shows only what a teacher needs today.	Simple class and section structures.	**INITIAL MVP / STAGE 1**
Simple operational analytics	Helps administrators see attendance and collection status.	Keep deterministic; avoid predictive claims.	**INITIAL MVP / STAGE 1** (basic); **PLANNED LATER (V1)** (management dashboards)
Selected offline workflows	Continues critical work during outages.	Intermittent connectivity is a realistic constraint; prevalence UNKNOWN.	**INITIAL MVP / STAGE 1** (attendance POC only)

> **What not to copy**
> Do not blindly copy international ERP practices. Practices should be adopted only where they solve a demonstrated Indian workflow problem, and they should be simplified rather than reproduced in full.

8. Customer Problems & Public Evidence

8.1 Nature of the evidence

The themes below recur across customer reviews, app-store feedback, public discussions, institutional material and procurement-style documents. They are **recurring, evidence-backed themes**. They are not universal conditions, and no percentage of schools experiencing any of them is claimed. The evidence supports the existence of each problem area; in most cases it does not establish its magnitude.

8.2 Theme summary

Theme	Evidence status / interpretation	Magnitude
Hybrid ERP + Excel + WhatsApp + paper workflows	Repeated public evidence that fragmentation persists even where software exists.	[UNKNOWN]
Communication overload	Customer discussions and product positioning repeatedly address large communication volumes.	[UNKNOWN]
Mobile / app usability issues	Public reviews include usability, reliability and workflow-friction complaints.	[UNKNOWN]
Support delays	Customer reviews and discussions identify support responsiveness as an important consideration.	[UNKNOWN]
Connectivity / offline needs	Public evidence supports the existence of constraints.	[UNKNOWN] (prevalence)
PTM coordination	Scheduling and booking is a recurring workflow opportunity.	[UNKNOWN]
Paper permissions / forms	Digital forms and consent are recurring opportunities.	[UNKNOWN]
Multi-campus reporting	Appears in enterprise / chain-school requirements and vendor capabilities.	[UNKNOWN]
Report-card customisation	Appears as an operational requirement across school contexts.	[UNKNOWN]
Integration gaps	Migration and integration concerns recur in procurement and customer discussions.	[UNKNOWN]
Fee reconciliation burden	Public evidence supports the problem area.	[UNKNOWN]
Teacher administrative friction	Recurring across attendance, marks and communication workflows.	[UNKNOWN]

8.3 Workflow analysis by theme

Hybrid workflows: ERP plus Excel, WhatsApp and paper

A school that owns an ERP can still run attendance summaries in Excel, announcements over WhatsApp groups and permissions on paper. The recurring evidence suggests that adoption of a system does not automatically remove these parallel processes. [ANALYST OBSERVATION] The likely mechanisms are that the system does not fit a real workflow well enough (so staff route around it), that migration was partial, or that particular tasks (parent messaging, permissions) were never covered. Each parallel process creates duplicate data entry and a second source of truth. **ZenCro implication:** design one coherent workflow where the same record feeds attendance, parent visibility and reports, and where Excel is an import and export format rather than a working environment.

Communication overload and targeting

Public discussions and product positioning both refer to heavy communication load on schools and on parents. The unresolved question is whether the problem is volume, poor targeting (messages reaching people who do not need them), fragmentation across channels, or all three. [ZENCRO HYPOTHESIS] Precise audience targeting (all parents, class, section, teachers, students) and a single place where announcements live may reduce noise. **Validation need:** how schools currently decide who receives what, and through which channel parents actually read.

Mobile usability and support responsiveness

App-store and review-platform evidence includes complaints about usability, reliability and friction, and customer material repeatedly identifies support responsiveness as significant. Reviews are self-selected and cannot show how common these issues are. [ZENCRO HYPOTHESIS] For a small team, a mobile-friendly web experience with a small surface area is more maintainable than several native apps, and a narrow scope is easier to support well. **ZenCro implication:** mobile-friendly web before native apps; support model designed alongside the product (Section 25).

Connectivity and offline needs

Public evidence supports the existence of connectivity constraints, but not their prevalence or duration. Teacher attendance is the most time-critical and frequent classroom task, which makes it the natural first candidate for local capture. **ZenCro implication:** a narrow, testable offline attendance proof of concept rather than a claim that the entire ERP works offline (Section 21).

PTM coordination and paper permissions

Parent-teacher meeting scheduling and paper-based permission and consent are both repeatedly visible as workflows that are manual, coordination-heavy and easy to lose or forget. Both are also self-contained and relatively simple to model, which makes them attractive early differentiating features. [ZENCRO HYPOTHESIS] Digital PTM booking and basic forms could deliver visible value with limited scope. **Validation need:** how schools run PTMs today and which forms are actually collected.

Fee reconciliation

Fee handling is a commoditized capability, but public evidence supports the burden of reconciling collections, dues and receipts. The magnitude is not established. **[ZENCRO HYPOTHESIS]** Clear dues after partial payments, immutable receipts and a daily collection view could reduce the reconciliation effort. This is a plausible execution advantage rather than a proven one.

Multi-campus reporting, report cards and integration gaps

These themes appear more in enterprise, chain-school and procurement material. Report-card customisation appears across contexts and is also a switching barrier (Section 9.3). Multi-campus depth is therefore treated as later work, unless target customers require it earlier. Integration gaps and migration concerns recur in procurement discussions, which supports treating migration and export as a first-class feature.

8.4 Traceability from problem theme to proposed response

Problem theme	Proposed response	Status	Validation needed
Hybrid workflows	Single coherent workflow; import / export instead of Excel working copies.	**INITIAL MVP / STAGE 1**	Which parallel processes matter most to schools?
Communication overload	Audience-targeted in-app announcements.	**INITIAL MVP / STAGE 1**	Which channels parents actually use?
Mobile usability	Mobile-friendly web; teacher-first dashboard.	**INITIAL MVP / STAGE 1**	Which devices do teachers use?
Support delays	Support model designed with the product.	**INITIAL MVP / STAGE 1** (operational)	What support level do schools expect?
Connectivity	Offline attendance capture with sync status.	**INITIAL MVP / STAGE 1** (POC-gated)	How often, and where, does connectivity fail?
PTM coordination	PTM slots and booking.	**INITIAL MVP / STAGE 1**	How are PTMs run today?
Paper forms / consent	Basic digital forms.	**INITIAL MVP / STAGE 1**	Which forms are collected and by whom?
Fee reconciliation	Deterministic fee rules and basic reconciliation.	**INITIAL MVP / STAGE 1**	How do accountants reconcile today?
Report-card customisation	Bounded configurable report card.	**INITIAL MVP / STAGE 1**	Which formats are required?
Integration gaps	CSV / Excel exports first; integrations later.	**PLANNED LATER (V1)**	Which integrations are needed first?
Multi-campus reporting	Deferred.	**PLANNED LATER (V1)** / **FUTURE / VALIDATION-DEPENDENT**	Do target customers operate multiple campuses?

Not claimed

No statement in this document asserts that every school experiences these problems, that a given proportion of schools does, or that ZenCro solves them. The mapping above states design intent, not results.

9. Buyer Journey & Procurement

Exact buying behaviour is not statistically validated. The picture below is drawn from procurement-style documents, institutional material and customer discussions, and should be read as a working model **[ZENCRO HYPOTHESIS]** to be tested in prospective-school conversations.

9.1 Who may be involved in a purchase

Stakeholder	Plausible role in the decision	Status
School management	Approves spend and vendor choice; concerned with cost, control and reporting.	[ZENCRO HYPOTHESIS]
Principal	Sponsors operational fit; concerned with academic workflow and staff acceptance.	[ZENCRO HYPOTHESIS]
Administrators	Daily users of fees, records and reports; strong influence on evaluation.	[ZENCRO HYPOTHESIS]
Management committees	May be involved depending on school type and size.	[ZENCRO HYPOTHESIS]
Other decision-makers	Depending on school type and size.	[UNKNOWN]

Who holds the decisive vote, and how this differs by school type and size, is UNKNOWN.

9.2 Potential purchase triggers

- Opening a new school.
- Dissatisfaction with the current ERP.
- Growing administrative workload.
- Fee problems.
- Reporting problems.
- Communication problems.
- A need to centralise records.
- A wish to replace paper and manual workflows.

9.3 Switching barriers

Barrier	Why it matters	ZenCro mitigation (proposed)
Migration	Moving records and history is effortful and risky.	Migration treated as a product feature: templates, validation, preview, clear errors.
Training	Staff must relearn workflows.	Simple role-specific dashboards; short training material.
Report-card customisation	Existing formats may be tightly tuned.	Bounded customisation in Stage 1; broader formats later.
Operational disruption	Switching mid-year can disrupt fees and attendance.	Controlled pilot; guidance on switch timing.
Support	Uncertain support quality raises perceived risk.	Explicit support model (Section 25).
Hidden costs	Modules, integrations or services priced separately.	Transparent packaging once validated (Section 26).
Data ownership	Schools worry about being locked in.	Strong data ownership and export posture.
Export limitations	Incomplete exports make leaving hard.	Export for students, attendance, fees and results in Stage 1.
Implementation risk	Delays and incomplete setup undermine adoption.	Staged rollout; implementation duration remains UNKNOWN until pilots.

10. Market Gap Analysis

This analysis classifies each area by how commoditized it appears and where execution differences may still matter. It deliberately avoids the phrase "market gap" for anything that is merely plausible: the research does not establish a single uncontested gap.

10.1 Area-by-area interpretation

Area	Research interpretation	Implication for ZenCro
Core SIS	Highly commoditized	Necessary; not a differentiator. Focus on clean import and export.
Attendance	Highly commoditized; execution and offline reliability may still matter	Excellent teacher UX and offline POC; do not market attendance itself as unique.
Fees	Commoditized, but reconciliation and operational clarity may be differentiators	Invest in deterministic rules, receipts and basic reconciliation.
Exams / report cards	Common capability; customisation and usability remain relevant	Bounded, configurable report cards; avoid over-promising board formats.
Parent / teacher portals	Common capability; UX and workflow quality matter	Simple, mobile-friendly web.
Communication	Common capability; targeted, unified workflows are potential opportunities	Audience-targeted announcements.
Digital forms / consent	Potential execution opportunity	Basic forms in Stage 1.
PTM scheduling	Potential workflow opportunity	PTM booking in Stage 1.
Migration / export	Potential switching and onboarding differentiator	Treat as a product feature.
Integrations	Potential practical differentiator	Planned later, prioritised by demand.
AI / predictive analytics	Low-confidence opportunity; defer until validated	**FUTURE / VALIDATION-DEPENDENT**. Not in MVP.

10.2 The combined proposition

[ZENCRO HYPOTHESIS] The more defensible opportunity is a **combination** of workflow quality, reliability, portability and India-specific operational fit, rather than any single missing feature. The combination is credible only if the core operating loop is executed consistently well.

10.3 What would weaken this view

- Prospective schools report that their current system already handles these workflows well.
- Schools do not distinguish between products on usability, or choose almost entirely on price.
- Migration and export are not, in practice, a switching concern for target schools.
- Established vendors ship comparable improvements before ZenCro reaches the market.
- The target segment has no willingness to pay at a level that supports the support model required.

11. Product Strategy

11.1 Product direction

An India-focused K-12 school management and operations platform designed to reduce administrative friction across core school workflows. The product should be India-focused, K-12, workflow-oriented, simple, mobile-friendly, reliable, portable and maintainable.

11.2 Design principles

Principle	What it means in practice	Example decision
Simple enough for teachers and administrators	Use without heavy training; few steps for frequent tasks.	Attendance entry as an exception-based, single-screen task.
One coherent workflow	Modules share one data model rather than being disconnected.	Attendance, parent view and reports draw on the same records.
Strong data ownership and export	Schools can leave with their data.	Export for students, attendance, fees and results in Stage 1.
Migration support	Onboarding is a product feature.	Templates, validation, preview and clear error messages.
Reliable core before advanced automation	Trust in the operating loop precedes AI and automation.	Defer AI; harden attendance and fees first.
Mobile-friendly web before native apps	A small team maintains one client.	Responsive web app; native apps **FUTURE / VALIDATION-DEPENDENT**.
India-specific fit without lock-in	Support Indian operations without hard-wiring one board or provider.	Configurable report cards; provider-agnostic payment approach.
Deterministic rules for fees, attendance and grades	Core records follow explicit business logic.	AI may assist but never controls core records.

11.3 Scope discipline: rules for adding scope

"More features" is not a strategy in a market where features are already widely offered. Any proposed addition to Stage 1 should pass these gates, and failure on any of them means the item is recorded as **PLANNED LATER (V1)** or **FUTURE / VALIDATION-DEPENDENT** rather than added:

- Does it belong to the core operating loop of a school (records, attendance, fees, results, communication, parent access)?
- Is it supported by public evidence or by direct customer validation, rather than by feature envy?
- Can it be built and supported by a small team without new infrastructure?
- Does it depend on integrations or data that may not exist at the pilot schools?
- Would a pilot school be unable to run its core workflows without it?

12. Target Segment Hypothesis

> **Status**
> The target segment is a working hypothesis. It has not been validated through interviews, surveys or sales conversations. It must not be presented as a validated target market.

12.1 The working hypothesis

[ZENCRO HYPOTHESIS] Small-to-medium schools, including schools with hybrid digital and manual workflows, may be a useful starting segment. A range of roughly 150-500 students was explored as a working hypothesis. This range is a conversation aid used to keep early product scope realistic; it is not a market definition, and no evidence in the research establishes that schools of this size share buying behaviour, budgets or needs.

12.2 Characteristics of a good early pilot school (proposed)

Characteristic	Reason	Status
Single campus	Avoids multi-campus complexity in Stage 1.	**[ZENCRO RECOMMENDATION]**
Current workflows spread across tools (ERP, Excel, WhatsApp, paper)	Provides a clear before-and-after workflow comparison.	**[ZENCRO RECOMMENDATION]**
Willing to share data samples for import testing	Migration is a product feature and needs realistic data.	**[ZENCRO RECOMMENDATION]**
A responsible decision-maker available for regular feedback	Fast learning is more important than scale at the pilot stage.	**[ZENCRO RECOMMENDATION]**
Ordinary, not ideal, connectivity	Makes the offline attendance POC meaningful.	**[ZENCRO RECOMMENDATION]**

12.3 What must be validated

Question	Why it matters
Do schools in this range actually purchase paid software, and who decides?	Determines the buyer and sales approach.
What do they currently use, and what do they dislike?	Determines differentiation.
What is their willingness to pay and preferred pricing basis?	Determines the business model ([UNKNOWN]).
How do their boards, state requirements and report-card formats affect needs?	Determines report-card and reporting scope.
Do they require integrations (payments, accounting, messaging) from day one?	Determines integration sequencing.

13. Product Scope & Stakeholders

13.1 Scope summary

In scope	Out of scope
K-12 school management and operations. India-focused product fit. The Stage 1 operating loop (Section 14). Later planned and future features as separately labelled tiers.	College and university ERP. Items on the Stage 1 exclusion list (Section 16). Any assumption that ZenCro is already built or already deployed.

13.2 Primary roles in the initial product

Role	Primary purpose in the product	Typical tasks (proposed)
School Admin	Configures and operates the school account.	School setup, users and roles, student records, imports and exports, announcements, PTM and form set-up.
Principal	Oversight and communication.	Review dashboards, attendance and results, publish announcements.
Teacher	Daily classroom operations.	Mark attendance, enter marks, view class information, PTM slots.
Accountant	Fee administration.	Fee structures, assignments, record payments, receipts, outstanding and collection reports.
Parent	Visibility and response.	View child's attendance, fees, receipts and results; announcements; book PTM slots; complete forms.

Student is not a login role in the initial product; audience targeting can address students as a recipient group for announcements, but the initial product's access model is built around the five roles above.

13.3 Proposed role-to-module access matrix

[ZENCRO RECOMMENDATION] The matrix below is a starting proposal to be refined during design. It defines the intended default; enforcement is described in Sections 20 and 23. M = manage (create / edit), V = view, VO = view own or own scope, E = enter data within scope, - = no access.

Module	Admin	Principal	Teacher	Accountant	Parent
School setup	M	V	-	-	-
Users and roles	M	V	-	-	-
Students	M	V	V (own classes)	V (fee-related)	VO (own child)
Attendance	M	V	E (own classes)	-	VO (own child)
Fees and receipts	V	V	-	M	VO (own child)
Exams and results	M	V	E (own subjects)	-	VO (own child)
Announcements	M	M	M (own audiences)	-	V
PTM	M	M	V / set slots	-	Book
Forms	M	M	M (as permitted)	-	Respond
Import and export	M	V (exports)	-	Export (fees)	-

14. Initial MVP / First Stage

INITIAL MVP / STAGE 1 Proposed product features, not claims that they are already implemented.

14.1 Purpose and boundary

The first stage is deliberately limited to the workflows that form the core operating loop of a school: who the students are, whether they attended, what they owe and have paid, how they performed, what the school communicated, and how parents engage. Each capability is scoped to its simplest reliable form. Anything beyond that is planned for later or excluded.

14.2 Module summary

#	Module	Purpose	Primary roles
1	School Setup	Configure the school, academic year, classes, sections, subjects, users, roles and permissions.	Admin
2	Student Management	Maintain student and guardian records, status and basic documents.	Admin, Principal, Teacher
3	Attendance	Record, correct and report attendance; parent visibility; selected offline capture.	Teacher, Admin, Parent
4	Fees	Fee structure, assignment, dues, payments, receipts, outstanding and collection reports, basic reconciliation.	Accountant, Admin, Parent
5	Exams & Results	Exams, subjects, marks, grading, results and report cards.	Teacher, Admin, Parent
6	Communication	Announcements with audience targeting.	Admin, Principal, Teacher
7	Parent Dashboard	Single parent view of child, attendance, fees, receipts, results, announcements, PTM and forms.	Parent
8	PTM	Create PTMs, teacher slots, parent booking and confirmation.	Admin, Teacher, Parent
9	Basic Forms	Permission, consent, event participation and acknowledgement with parent responses.	Admin, Parent
10	Import & Export	CSV / Excel upload with validation and preview; exports of core data.	Admin
11	Dashboards	Admin, Teacher and Parent dashboards.	All roles

14.3 Dependencies and build logic

School Setup and the access model are prerequisites for everything else. Student records feed attendance, fees, exams and parent access. Attendance, fees and results feed the dashboards and the parent view. Communication, PTM and forms sit on top of the audience model (who is a parent of which class and section). Import and export cut across all modules and should be built as the data model stabilises. The recommended build sequence is in Section 24.

14.4 What "done" means for the MVP (qualitative)

Success criteria for the first stage are qualitative because ROI and adoption benchmarks are UNKNOWN and must not be invented. A pilot school should be able to:

- Set up its academic structure and users and load its students from a spreadsheet.
- Take attendance daily with teacher effort the pilot school judges acceptable, with parents able to see it.
- Assign fees, record full and partial payments, issue receipts and see outstanding and collection reports.
- Enter marks and produce report cards in an agreed format.
- Communicate to targeted audiences, schedule a PTM and collect at least one type of parent response digitally.
- Export its core data at any time.

Quantitative thresholds should be defined jointly with pilot schools and recorded after the pilot, not set in advance from assumptions.

15. MVP Functional Modules

INITIAL MVP / STAGE 1 Functional specification for the initial build. Descriptions of behaviour are ZenCro proposals to be validated.

15.1 School Setup

Scope items	Behaviour and business rules (proposed)
• Create school • Academic year • Classes • Sections • Subjects • Users • Roles • Permissions	• Each school is a separate tenant; every record carries the school identity. • One academic year is active at a time, with earlier years retained for reference. • Classes contain sections; subjects can be attached to classes. • Users are invited to a school and assigned one or more roles. • Permissions follow the role matrix in Section 13.3 and are enforced in the database (Section 20).

Design note: Setup should be completable by a school admin with a clear checklist and reversible steps. Open question: how flexible must class and section structures be for different boards?

15.2 Student Management

Scope items	Behaviour and business rules (proposed)
• Student profile • Student / admission ID • Name, DOB, gender • Class / section • Parent / guardian • Contact details • Student status • Basic document upload	• Admission ID is unique within a school. • A student can have one or more linked guardians; a guardian may link to more than one student. • Status values (for example active or left) drive which lists and reports include the student. The list is configurable in design. • Documents are stored privately and accessible only through role-checked access.

Design note: Changes to sensitive fields should be traceable. Open question: which additional profile fields do target schools need for their own reporting?

15.3 Attendance

Scope items	Behaviour and business rules (proposed)
• Daily / class attendance • Teacher entry • Edit / correction • Attendance percentage • Reports • Parent visibility • Basic notification • Selected offline attendance	• A teacher opens a class and section and sees the roster for the day. • One attendance record exists per student per date (and per session if the school uses more than one); duplicates are prevented by a database constraint. • Corrections keep a history: who changed what, when and why, subject to an admin-configurable edit policy. • Percentages are calculated deterministically from stored records. • Parents see their own child's attendance only. • Basic notification is an in-app notification in Stage 1; external channels are later. • Offline capture is limited to attendance entry and is gated by the proof of concept (Section 21).

Design note: Speed of entry is the main usability target. Open question: whether the default state should be present or absent, and how many sessions per day schools use.

15.4 Fees

Scope items	Behaviour and business rules (proposed)
• Fee categories • Fee structure • Fee assignment • Student dues • Record payment • Partial payment • Receipt • Outstanding report • Collection report • Basic reconciliation	• Fee categories are defined once and combined into fee structures by class and academic year. • Assignment creates dues for each student from the applicable structure. • Payments are recorded against dues, allowing partial payments; the remaining balance is calculated from records, not typed by hand. • Receipts are numbered sequentially per school and academic year and are not edited after issue; corrections are made through reversal or adjustment records. • Outstanding and collection reports are available by class, date and payment mode. • Money is stored in an exact numeric form, never as floating-point values. • Online payment gateway integration is planned later rather than building a custom payment system.

Design note: Basic reconciliation is proposed to mean internal consistency: dues, payments and receipts agree, and a daily collection summary can be closed by the accountant. Bank or gateway matching is a later capability. Open question: concessions, discounts and late fees (not listed in the initial scope) need a decision.

15.5 Exams & Results

Scope items	Behaviour and business rules (proposed)
• Create exam • Subjects • Enter marks • Grading • Results • Report card	• An exam is defined per class and academic year with its subjects and maximum marks. • Teachers enter marks only for their assigned subjects and sections. • A grading scale is configurable at school level; results are computed deterministically. • Report cards are generated from stored results using a bounded set of configurable templates. • Results become visible to parents only when published by an admin.

Design note: Report-card customisation is a recurring evidence theme and a switching barrier, so the scope of configurability is a key design decision. Board-specific formats beyond the initial core are FUTURE / VALIDATION-DEPENDENT.

15.6 Communication

Scope items	Behaviour and business rules (proposed)
• Announcements • Audience targeting • Publish / view announcements	• Audience targeting options: all parents, class, section, teachers or students. • Announcements are published in-app and viewable in the relevant dashboards. • Targeting is resolved from enrolment and guardian relationships so that messages reach the correct parents.

Design note: External email, SMS, WhatsApp and push integrations are later (PLANNED LATER (V1)). Open question: whether in-app announcements alone can reach parents who mainly use WhatsApp.

15.7 Parent Dashboard

Scope items	Behaviour and business rules (proposed)
• Child • Attendance • Fees • Receipts • Results • Announcements • PTM • Forms	• A parent sees each linked child; a parent with more than one child can switch between them. • Data shown is limited to that parent's linked children through database-level access rules. • Fees show dues, payments and receipts; online payment initiation is later. • Forms and PTM slots appear as clear pending actions.

Design note: Designed to work well on a phone browser. Open question: parent authentication approach and account recovery for parents without email.

15.8 PTM

Scope items	Behaviour and business rules (proposed)
• Create PTM • Date • Teacher slots • Parent slot selection • Booking confirmation	• An admin creates a PTM event with a date and the participating classes. • Teachers define or receive time slots; a slot can be booked once (double-booking is prevented in the database). • A parent selects a slot for a teacher; a confirmation is shown to the parent and teacher. • Cancellation and rescheduling rules are set during design.

Design note: Slot rules (one slot per parent-teacher pair, maximum slots per parent) are open design questions to settle with pilot schools.

15.9 Basic Forms

Scope items	Behaviour and business rules (proposed)
• Permission • Consent • Event participation • Acknowledgement • Parent response	• An admin creates a form, chooses the audience and publishes it. • A parent accepts, rejects or submits a response; the response records who responded and when. • A form version is preserved so that responses relate to the exact text shown. • Responses can be viewed and exported by authorised staff.

Design note: Whether a digital response satisfies a given legal or institutional requirement is for legal review; the product records the response, it does not make the legal determination (Section 23.2).

15.10 Import & Export

Scope items	Behaviour and business rules (proposed)
• CSV / Excel upload • Validation • Error display • Preview • Confirm import • Export students • Export attendance • Export fees • Export results	• Upload, then validation, then a row-by-row error display, then a preview of what will be created or updated, then an explicit confirmation. • Nothing is written to school data until the admin confirms. • Exports cover students, attendance, fees and results in CSV or Excel format. • Imports and exports are logged.

Design note: Migration is treated as a product feature (Section 24.2). Open question: which additional entities (guardians, historical fees) must be importable in Stage 1.

15.11 Dashboards

Scope items	Behaviour and business rules (proposed)
• Admin dashboard • Teacher dashboard • Parent dashboard	• Admin: students, classes, attendance, fee collection and outstanding, exams, PTM, announcements. • Teacher: classes, attendance, pending marks, PTM, announcements. • Parent: child, attendance, fees, results, announcements, PTM and forms.

Design note: Dashboards use deterministic, simple summaries. Predictive or AI-generated insights are not part of Stage 1.

16. Explicit Stage 1 Exclusions

NOT PART OF INITIAL MVP / STAGE 1 Scope-control decisions. These items must not expand the first build unless a new requirement is explicitly approved.

The exclusions protect the first stage from turning into an oversized ERP. They are decisions about sequencing and focus, not judgements that the items lack value.

16.1 Excluded product modules

Excluded item	Status	Reason for exclusion
Library management	**NOT PART OF INITIAL MVP / STAGE 1**	Outside the core operating loop; adds a separate data model and workflow.
Hostel management	**NOT PART OF INITIAL MVP / STAGE 1**	Applies to a subset of schools; specialised workflow.
HR and payroll	**NOT PART OF INITIAL MVP / STAGE 1**	Substantial compliance and accuracy demands; different buyer and user group.
Staff leave management	**NOT PART OF INITIAL MVP / STAGE 1**	Depends on HR structures not in Stage 1.
Advanced timetable engine	**NOT PART OF INITIAL MVP / STAGE 1**	Complex constraint solving; better informed by real usage first.
Full LMS	**NOT PART OF INITIAL MVP / STAGE 1**	A large product in its own right; not needed to run the operating loop.
Homework management	**NOT PART OF INITIAL MVP / STAGE 1**	Adds daily teacher and parent workflows; planned later in basic form.
Advanced admissions CRM	**NOT PART OF INITIAL MVP / STAGE 1**	Separate funnel workflow; planned later.
Inventory management	**NOT PART OF INITIAL MVP / STAGE 1**	Separate operational domain.
School accounting	**NOT PART OF INITIAL MVP / STAGE 1**	Dedicated accounting depth; Stage 1 covers only fee collection records and exports.
Complex multi-campus architecture	**NOT PART OF INITIAL MVP / STAGE 1**	Adds tenancy and reporting complexity; revisit if target customers require it.
AI chatbot	**NOT PART OF INITIAL MVP / STAGE 1**	Value not validated; risk to core-record integrity.
Predictive analytics	**NOT PART OF INITIAL MVP / STAGE 1**	Requires data quality and history that do not exist yet.
Native Android / iOS applications	**NOT PART OF INITIAL MVP / STAGE 1**	Mobile-friendly web first; a small team should maintain one client.

16.2 Excluded technical components

Excluded component	Status	Reason for exclusion
Microservices	**NOT PART OF INITIAL MVP / STAGE 1**	Operational overhead not justified by first-stage requirements.
Kubernetes	**NOT PART OF INITIAL MVP / STAGE 1**	Infrastructure complexity without a scale requirement.
Separate backend server	**NOT PART OF INITIAL MVP / STAGE 1**	Next.js server-side routes and actions are sufficient initially.
Redis	**NOT PART OF INITIAL MVP / STAGE 1**	No caching or queueing requirement established.
Dedicated message queue	**NOT PART OF INITIAL MVP / STAGE 1**	No asynchronous workload that requires it in Stage 1.
Vector database	**NOT PART OF INITIAL MVP / STAGE 1**	Only relevant to AI retrieval features that are not in Stage 1.
Complex event-driven architecture	**NOT PART OF INITIAL MVP / STAGE 1**	A modular monolith with a relational database is simpler to build and reason about.

16.3 Change-control rule

An excluded item may enter scope only through an explicit requirement approval that states the customer evidence behind it, what will be removed or deferred to make room, and its effect on the pilot. Requests arriving during the pilot are logged and considered for **PLANNED LATER (V1)** planning rather than absorbed into Stage 1.

17. Later / V1 Planned Features

PLANNED LATER (V1) Planned for after the core MVP is stable. These are not implemented and are not part of the first-stage build.

Area	Planned later feature	MVP foundation it builds on	Depends on
Admissions	Structured admissions workflow, application tracking and document collection	Student records, document upload	Customer evidence of admissions needs
Homework / learning	Homework, classwork, teacher uploads, basic learning resources	Classes, subjects, teacher dashboard	Storage and notification design
Timetable	More advanced timetable creation; teacher and class conflict handling	Classes, sections, subjects, teachers	Real scheduling requirements from pilot schools
Communication	Email, SMS, WhatsApp, push integrations; richer message templates	In-app announcements and audience targeting	Provider choices; cost and consent model
Payments	Online payment gateway integration; stronger reconciliation	Fee structures, receipts, basic reconciliation	Provider choice; legal and operational review
Reports	More customisable reports and management dashboards	Core reports and dashboards	Data quality after real usage
Documents	Expanded document management and digital forms	Basic document upload, basic forms	Retention and access policy
Migration	Improved migration tools, templates and guided onboarding	Import and export	Pilot learnings on real source data
Integrations	Accounting, payment and communication integrations based on customer demand	Exports; stable data model	Validated demand
Offline	Broader offline workflows if the attendance POC proves reliable	Offline attendance capture	POC success
Multi-campus	Stronger multi-school / campus controls if target customers require them	Tenant model, RBAC	Customer requirement

18. Future / Validation-Dependent Features

FUTURE / VALIDATION-DEPENDENT Potentially valuable, but current public evidence is not strong enough to justify building them. Not committed features.

Feature	Why it may be valuable	Why it is deferred	What would justify it
AI-assisted summaries and drafting	Could reduce administrative writing effort.	Value unvalidated; data-handling implications.	Pilot demand; a safe, reviewable workflow.
AI administrative assistants	Could simplify navigation and routine queries.	Risk to core records if poorly bounded.	Stable core workflows and data quality.
Predictive analytics	Could flag attendance or performance patterns.	Requires sufficient historical data; higher risk of misinterpretation.	Sufficient real data and school appetite.
Advanced school performance analytics	Deeper insight for management.	Depends on reliable data over time.	Pilot history; validated management needs.
Wellbeing workflows	May support pastoral needs.	Sensitive data; needs expert and legal input.	Clear demand and governance model.
Advanced automation across modules	Could reduce manual steps.	Complexity; correctness risk.	Evidence of repetitive workflows worth automating.
Native mobile apps	Possibly better device integration.	Maintenance cost; unproven benefit over mobile web.	Evidence that mobile web is insufficient.
Large-scale multi-campus administration	Serves school groups.	Complex tenancy and reporting.	Customer demand and a target segment shift.
Sophisticated accounting / finance integrations	Reduces double entry for schools using external tools.	Demand varies; integration effort high.	Validated demand for specific tools.
Specialised board / state workflows	Deeper fit for particular regions.	Varying requirements; high maintenance.	Concentrated customer demand.

18.1 AI guardrail

AI must not be the source of truth
AI should assist rather than control core records. Fees, attendance, grades, payments and other core school records must remain deterministic database and business-logic workflows. Any AI output that touches a record must be reviewable and reversible by a human.

18.2 Preconditions before any AI work

- Core workflows are stable and data quality is demonstrably good.
- A clear use case validated with schools, with a defined human-review step.
- Legal and privacy review of any student or family data sent to an external AI service (the initial direction names the Google Gemini API as a possible later option). Whether and how such data may be processed is a DPDP workstream question, not a technical detail.
- Cost and reliability review of the AI service.

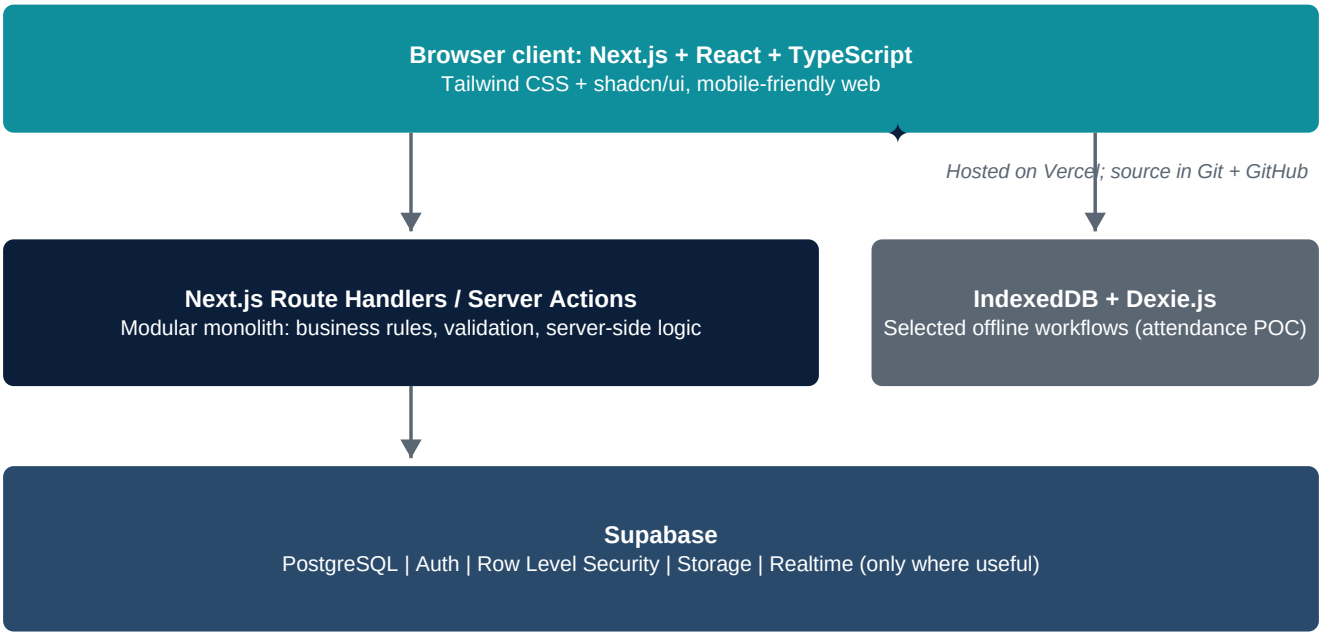
19. Technical Architecture

The technical direction is intentionally simple: one main web application, one primary backend layer inside that application, and Supabase for managed database, authentication and storage. This keeps development easier for a small team, reduces infrastructure to operate, and matches the discipline of the product scope. The stack below is the recommended first-stage direction **[ZENCRO RECOMMENDATION]**.

19.1 Recommended first-stage stack

Layer	Recommended technology	Purpose
Frontend	Next.js + React	Web application and dashboards
Language	TypeScript	Shared type safety across the application
Styling	Tailwind CSS	Fast, consistent UI development
UI components	shadcn/ui	Reusable accessible components
Backend	Next.js Route Handlers / Server Actions	Business logic and API endpoints without a separate backend server
Architecture	Modular monolith	Simple deployment and easier maintenance
Database	Supabase PostgreSQL	Primary relational school data
Authentication	Supabase Auth	User sign-in and session management
Authorization	RBAC + Supabase Row Level Security	Role and school-level data isolation
File storage	Supabase Storage	Documents and generated / uploaded files
Realtime	Supabase Realtime, only where useful	Selected live updates
Offline	IndexedDB + Dexie.js (selected workflows)	Local attendance capture and later synchronisation
AI	Google Gemini API (later)	Optional drafting / summarisation after MVP
Deployment	Vercel	Web app deployment
Version control	Git + GitHub	Source control and collaboration
Development	VS Code + Antigravity	Implementation and assisted development

19.2 Architecture at a glance



19.3 Modular monolith structure

A modular monolith means a single deployable application whose code is organised into clear domain modules with explicit boundaries. The proposed domain modules mirror the MVP modules: school and access, students, attendance, fees, exams, communication, PTM, forms, documents, and import / export. Modules communicate through defined service functions rather than reaching into one another's data access code. This preserves the option to extract a service later without paying that complexity now.

- **Server-side by default:** mutations that change core records (attendance, payments, marks) run in Server Actions or Route Handlers where validation and business rules are applied consistently.
- **Business rules in one place:** fee calculations, attendance percentages and grading rules are implemented as tested server-side functions, not scattered through UI code.
- **Validation at the boundary:** all input is validated and sanitised on the server, regardless of client-side checks.
- **Types shared end-to-end:** TypeScript types for entities and payloads reduce integration errors.

19.4 Why not a separate backend or heavier infrastructure (yet)

Component	Decision	Reason	Condition to revisit
Separate Node / Express backend	**NOT PART OF INITIAL MVP / STAGE 1**	First-stage requirements do not justify the operational cost of a second deployable.	A system boundary or scale need that Next.js server code cannot meet.
Microservices	**NOT PART OF INITIAL MVP / STAGE 1**	Adds distributed-systems complexity and operating burden.	Independent scaling or team boundaries that a monolith cannot serve.
Kubernetes	**NOT PART OF INITIAL MVP / STAGE 1**	No scale or deployment requirement that Vercel and Supabase do not already cover.	A hosting requirement that managed platforms cannot meet.
Redis	**NOT PART OF INITIAL MVP / STAGE 1**	No demonstrated caching or session need.	Measured performance problems that caching would solve.
Dedicated message queue	**NOT PART OF INITIAL MVP / STAGE 1**	No asynchronous workload requires it in Stage 1.	Reliable bulk notification or long-running jobs after integrations are added.
Vector database	**NOT PART OF INITIAL MVP / STAGE 1**	Only relevant to AI retrieval features that are not in Stage 1.	Validated AI use case that needs retrieval.
Complex event-driven architecture	**NOT PART OF INITIAL MVP / STAGE 1**	Relational transactions are simpler and sufficient.	Demonstrated need for decoupled event processing.

20. Supabase Architecture

20.1 Role of each Supabase capability

Capability	Role in the product	Design guidance (proposed)
PostgreSQL	Source of truth for schools, users, students, classes, attendance, fees, exams and related records.	Relational schema with foreign keys and constraints that enforce business invariants (for example one attendance record per student per date).
Auth	Sign-in and sessions for Admin, Principal, Teacher, Accountant and Parent.	Users are linked to schools and roles through membership records; parents are linked to students through guardian records.
Row Level Security (RLS)	Enforces school / tenant isolation and role-aware access at the database layer.	Enable RLS on every tenant-owned table; deny by default; explicit policies per role.
Storage	Student and school documents and generated files.	Private buckets; object paths prefixed by school identity; access through short-lived signed access after a role check.
Realtime	Selective live updates where they improve the experience.	Use only where immediacy matters (for example a PTM slot being taken); not required across the application.
Database migrations	Keep schema changes version-controlled and repeatable.	Every schema change is a reviewed migration file applied through the same process in every environment.

20.2 Tenancy model

The proposed model is a shared database and schema with a school identity on every tenant-owned row, protected by RLS. This is the simplest model to operate for a small team. Its principal risk is a policy mistake that exposes one school's data to another; that risk is addressed by defence in depth (Section 23) and by automated cross-tenant tests. Alternatives such as schema-per-tenant or database-per-tenant add operational overhead and are not recommended for Stage 1 **[ZENCRO RECOMMENDATION]**.

20.3 Access rules by actor (conceptual)

Actor	Access rule (conceptual)
Admin / Principal / Accountant	Rows where the school identity matches one of the user's school memberships, filtered further by the permissions of the assigned role.
Teacher	Rows for the classes, sections and subjects the teacher is assigned to; can write attendance and marks only within that scope.
Parent	Rows connected to students to whom the parent is linked as a guardian; read-only for most data; write access limited to responses (form answers, PTM bookings).
Server-side processes	Use the privileged service credential only in server-side code for operations that legitimately bypass user-level rules, and apply their own authorisation checks first.

20.4 Keys and secrets

> **Security note**
> The service-role secret must remain server-side and must never be exposed to the browser. Client-side access uses appropriately restricted keys and relies on RLS. Secrets are stored in the hosting platform's secure configuration and are never committed to source control.

20.5 Operational considerations

- **Environments:** separate development, staging and production projects, so that testing never touches real school data.
- **Performance:** index the columns used by policies and common filters (school identity, academic year, class, student, date). Measure before optimising.
- **Backups and recovery:** capabilities depend on the chosen Supabase plan. Recovery objectives are a business decision to define and test before commercial deployment.
- **Limits and cost:** plan limits (database size, storage, connections) should be reviewed against expected pilot load **[UNKNOWN]** before launch.

21. Offline Strategy

Keep it narrow
Offline capability is a targeted engineering hypothesis, not a requirement to make the entire ERP offline. The first candidate is teacher attendance. A complete offline ERP would introduce significant synchronisation, conflict-resolution and data-integrity complexity and is outside the first stage.

21.1 Why attendance first

Attendance is frequent, time-critical, performed by a defined role in a defined location, and has a compact data shape. Public evidence supports the existence of connectivity constraints (prevalence UNKNOWN), and vendor offline claims exist in the market [VENDOR CLAIM]. That combination makes attendance a good place to test whether offline capture is feasible and valuable without committing the whole product.

21.2 Proposed behaviour

Step	Behaviour
1. Online save	When connected, attendance is saved directly to the server and confirmed to the teacher.
2. Offline save	When not connected, attendance is saved locally in IndexedDB using Dexie.js, marked as pending, and clearly shown as not yet on the server.
3. Synchronise	When connectivity returns, pending records are sent to the server automatically or on demand.
4. Show status	Every record and every class shows a status: pending, synced, or needs attention.
5. Avoid silent loss	A pending record is never discarded until the server has confirmed it; failures remain visible with a retry option.

21.3 Design principles for synchronisation

- **Idempotent submission:** each local record carries a client-generated identifier so that repeating a sync cannot create duplicates.
- **Server as authority:** the server validates each submission against roster, date and permissions; invalid records are flagged, not dropped.
- **Conflict handling:** if the same class and date has been changed elsewhere, the conflict is surfaced to a person for resolution rather than silently overwritten.
- **Minimal local data:** store only what attendance entry needs (the roster snapshot and pending entries), reducing exposure of student data on shared devices.
- **Clear communication:** the teacher can always tell whether a class's attendance has reached the server.

21.4 Proof-of-concept plan

Scenario	What it tests	Expected result
Take attendance with no connection, close the browser, reopen later	Persistence of pending records	Records remain available and marked pending
Regain connection with pending records	Sync and status display	Records reach the server; status changes to synced
Interrupt sync partway (connection drops during upload)	Idempotency and retry	No duplicates; no lost records; failure is visible
Edit the same class and date on two devices	Conflict handling	Conflict is flagged for a person to resolve
Session expires while offline	Authentication on reconnect	Teacher reauthenticates without losing pending records
Test on the actual devices and browsers teachers use	Real-world storage behaviour	Behaviour is confirmed or limitations are documented
Clear the browser data with records pending	Data-loss awareness	Limitation understood and communicated in guidance

Browser storage can be cleared by users or, in some conditions, by the browser itself. The POC must therefore include the specific devices and browsers used by teachers, and the outcome should be an evidence-based decision about how far to go, not an assumption.

21.5 Decision gate

Success is defined qualitatively: no silent data loss in the tested scenarios, clear status at all times, safe handling of conflicts, and acceptable teacher effort on real devices. If the POC succeeds, broader offline workflows can be considered individually as **PLANNED LATER (V1)** items. If it fails or the maintenance cost is high, Stage 1 keeps attendance online-only and the finding is recorded.

22. Data Model

The model below is a first-stage concept. Attributes and relationships are proposals for design review, not a final schema.

22.1 Entities

Entity	Purpose	Key attributes (proposed)	Key relationships
School	Tenant and school-level configuration	Name, contact details, settings	Parent of all tenant-owned data
Academic Year	Academic period	Label, start and end dates, active flag	Belongs to a school; scopes enrolments, fees, exams
User	Authenticated system user	Identity from Auth, name, contact	Member of schools through roles
Role	Permission grouping	Role name, permissions	Assigned to a user within a school
Student	Core student record	Admission ID, name, DOB, gender, status	Belongs to a school; has guardians and enrolments
Parent / Guardian	Student-linked parent information	Name, relationship, contact	Links to one or more students; may link to a user
Class / Section	Academic grouping	Class name, section name	Belongs to a school and academic year
Subject	Subject catalogue	Subject name, code	Attached to classes; used by exams
Enrollment	Student-to-class/section relationship	Student, class, section, academic year, dates	Links student to class and section per year
Attendance	Daily / class attendance records	Student, date, status, marked by, timestamps	Unique per student per date (and session)
Fee Structure	Fee categories and amounts	Category, amount, class, academic year	Used to create student fee assignments
Student Fee	Assigned dues and balances	Student, fee item, amount due	Links student to fee structure items
Payment	Recorded payments and receipts	Amount, date, mode, receipt number, recorded by	Applied to student fees; receipts are immutable
Exam	Assessment event	Name, class, academic year, dates	Has subjects and max marks
Marks / Result	Subject marks and grades	Student, exam, subject, marks, grade	Belongs to an exam and student
Announcement	Targeted school communication	Title, body, audience, publish time, author	Targets classes, sections or roles
PTM / Slot / Booking	Parent-teacher scheduling	Event date, teacher, slot time, booked by	Slot belongs to a PTM and teacher; booking links a parent and student
Form / Response	Digital permission / consent workflow	Form text and version, audience, response, respondent, time	Response links to form version, student and respondent
Document	Stored school/student files	File reference, type, owner record, uploader	Attached to a student or school record

22.2 Modelling conventions

- **Tenant identity on every row:** every tenant-owned table carries the school identity so that RLS policies are simple and uniform.
- **Stable identifiers:** opaque unique identifiers for all records; human-readable numbers (admission ID, receipt number) are separate, unique per school.
- **Timestamps and actors:** creation and update times, and who made the change, on important records.
- **Exact money:** amounts held in exact numeric form (for example integer minor units); never floating point.

- **Append-only where it matters:** receipts and attendance corrections are recorded as new entries or reversals rather than overwriting history.
- **Constraints as guardrails:** unique, foreign-key and check constraints enforce invariants at the database level.
- **Status over deletion:** student records use status values (not hard deletion) so history is preserved, subject to the retention and deletion process in Section 23.

23. Security & Privacy

School systems process sensitive information about children and families. Security and privacy are treated as architectural requirements from the start, not as later add-ons.

23.1 Security controls

Control	Purpose	Direction (proposed)
Role-based access control	Restrict actions by role.	Defined role matrix (Section 13.3), enforced server-side and at the database.
Supabase Row Level Security	Database-level tenant and role isolation.	Deny by default; explicit tested policies; RLS on all tenant tables.
School / tenant isolation	Prevent cross-school data exposure.	School identity on every row; automated cross-tenant tests.
Server-side handling of sensitive operations	Keep privileged logic away from the browser.	Payments, marks and imports processed in server code with authorisation checks.
HTTPS	Protect data in transit.	Enforced in production.
Input validation and sanitisation	Reduce injection and data-quality problems.	Server-side validation for all inputs; safe handling of uploaded files.
Secure secrets management	Prevent credential leakage.	Secrets in secure configuration; service credential never in client code.
Audit-friendly records	Traceability for important changes.	Log who changed attendance corrections, payments, marks and role assignments, and when.
Backup and recovery	Restore after failure or error.	Procedures appropriate to the chosen plan; restoration tested before deployment.
Retention, deletion and export processes	Support data governance and data ownership.	Defined and documented before commercial deployment.

23.2 DPDP and compliance: a workstream, not a claim

> **Compliance position**
> ZenCro must not claim to be DPDP compliant merely because it uses Supabase or Row Level Security. Compliance with data protection law is a product, process and legal workstream that requires ongoing review. Architecture supports compliance; it does not establish it.

India's Digital Personal Data Protection Act, 2023 is relevant to a product that processes children's and families' personal data, and it contains provisions specific to children's data. How it applies to school operations, how roles and responsibilities are allocated between school and ZenCro, and what processes are required should be determined with qualified legal counsel against the requirements in force at the time of deployment. The table below lists the workstreams to define, not the answers.

Workstream	Question to resolve	Function
Roles and responsibilities	How are responsibilities divided between the school and ZenCro for the data processed?	Legal, Product
Lawful basis and consent	What consents or notices are needed, and how are they captured and recorded?	Legal, Product
Children's data	What specific requirements apply to data about children, and how is this reflected in the product?	Legal, Product, Engineering
Retention and deletion	How long is data kept, and how is deletion handled, including backups?	Legal, Product, Engineering
Access and export requests	How does the school (and where relevant a parent) request access or export?	Product, Engineering
Incident response	What is the process, and who is notified, if a breach occurs?	Legal, Engineering
Vendors and sub-processors	What agreements and reviews cover Supabase, Vercel and any future AI or messaging provider?	Legal, Product

Workstream	Question to resolve	Function
Data location	Where is data hosted, and is that acceptable for schools and their obligations?	Legal, Engineering

23.3 Security process commitments before the pilot

- Automated tests for role and tenant isolation.
- A security review of authentication, authorisation, file access and secret handling.
- Documented backup and restore test.
- Documented incident response outline.
- Documented retention, deletion and export procedures.

24. Implementation & Migration

24.1 Recommended build sequence

The sequence below builds the foundation first (structure, identity, access control), then the daily-use workflows, then parent-facing and cross-cutting features, and ends with testing and a controlled pilot. Exit criteria are proposed quality gates, not time estimates; duration is not established by the research.

Step	Scope	Exit criteria (proposed)
1	Project setup and repository	Repository, environments and deployment pipeline working; coding standards agreed.
2	Supabase project and database schema	Core schema in migrations; constraints in place.
3	Authentication + RBAC + RLS	Roles enforced; tenant and role isolation tests pass.
4	School setup	Admin can configure year, classes, sections, subjects, users.
5	Student management	Student and guardian records; documents stored privately.
6	Attendance	Teacher entry, corrections, percentages, reports, parent view.
7	Fees	Structures, dues, payments, receipts, reports; rule tests pass.
8	Exams & results	Marks entry, grading, results, report card output.
9	Communication	Targeted announcements viewable by the correct audiences.
10	Parent dashboard	Parent view limited to linked children.
11	PTM + forms	Slot booking without double-booking; form responses recorded.
12	Import / export	Validated import with preview and confirm; exports for core data.
13	Testing + security review	Functional, role, tenant and security tests reviewed; issues resolved or accepted.
14	Deployment and controlled pilot	Pilot school live with monitoring, backups and support process in place.

24.2 Migration as a product feature

Migration is a switching barrier and a recurring concern in public evidence, so it should be designed as a product feature and not as a one-off service. The exact time to migrate depends on the school's source data and is not established by the research **[UNKNOWN]**.

Migration stage	What the product provides
1. Prepare	Downloadable templates with clear instructions; guidance on collecting data from Excel or paper sources.
2. Upload	CSV / Excel upload with file-level checks.
3. Validate	Field-level checks (required fields, formats, duplicates, unknown classes).
4. Review errors	Row-by-row errors with plain-language explanations so the school can fix and re-upload.
5. Preview	Summary of what will be created or updated before anything is saved.
6. Confirm	An explicit confirmation step; nothing is written before it.
7. Reconcile	Post-import counts and a report so the school can verify the result.
8. Export	Complete export of core data at any time.

24.3 Onboarding and change management

- Kick-off with the school to confirm structure, classes, roles and formats.
- Sample-data trial import before the full import.
- Role-specific short training for teachers, accountants, admins and parents.
- A defined support channel during the first weeks.
- A go-live plan that considers the academic calendar and any parallel running of old and new processes.
- Hidden costs and data ownership terms explained clearly in advance.

24.4 Controlled pilot

The pilot should involve a small number of schools that fit the criteria in Section 12.2, with defined goals, an agreed feedback cadence and explicit permission to change scope. The pilot is a learning exercise to validate product value, migration effort and support demands. It should not be presented as market validation of willingness to pay unless pricing is explicitly tested with those schools.

25. Support & Operations

Support quality may influence switching and retention. Customer evidence repeatedly identifies support responsiveness as a consideration. Exact SLA expectations and willingness to pay for premium support are unknown **[UNKNOWN]**. Any premium SLA is a business decision that must be validated with customers.

25.1 Support model options (to be validated)

Option	Description	Considerations
Self-service resources	Help articles, short guides, import templates.	Low cost; requires good documentation.
Ticket / email support	Tracked requests with a response process.	Enables measurement and follow-up.
Chat or phone support	Faster human contact.	Higher cost; expectation of availability during school hours.
Onboarding assistance	Guided setup and migration help.	May be charged as an implementation fee (Section 26).
Premium support tier	Priority or extended-hours support.	Demand and price UNKNOWN; must be validated.

26. Business Model Considerations

> **Pricing status**
> The research does not establish an exact willingness-to-pay range. No pricing is proposed in this document. Competitor pricing that is not public is recorded as NOT PUBLICLY DISCLOSED - QUOTE REQUIRED and is not estimated.

26.1 Candidate commercial structures

Model	Advantages (hypotheses)	Risks / open questions
Annual subscription per school	Simple to explain and forecast.	May not scale fairly with school size.
Per-student pricing	Scales with school size.	Schools may object to variable cost; needs a clear definition of a billable student.
Tiered plans by size or features	Allows packaging by size or capability.	Risk of confusing tiers or of holding back features schools consider basic.
Implementation / migration fee	Covers onboarding effort.	May act as a barrier; effort UNKNOWN.
Optional paid integrations	Aligns cost with demand.	Integration demand varies by school.
Optional premium support	Monetises higher service levels.	SLA expectations and willingness to pay UNKNOWN.

26.2 How pricing should be validated

- Explore pricing basis preferences with prospective schools (per school, per student, tiered) before choosing.
- Ask about current spending, who approves spend, and what would justify a change, without presenting a price as a fact.
- Test whether schools view migration help and support as included or extra.
- Compare the total cost picture (including hidden costs) that schools associate with current vendors.

- Record results as customer evidence and update the assumptions register (Section 28).

27. Product Roadmap

The roadmap is organised into stages rather than dates. Timing depends on team size, pilot learnings and validation results, none of which the research establishes.

Stage	Focus	Status / intent
Stage 1: MVP	School setup, student records, attendance, fees, exams / results, communication, parent dashboard, PTM, basic forms, import / export	**INITIAL MVP / STAGE 1** Build first
Stage 2: V1	Admissions, homework / basic learning, timetable improvements, payment and communication integrations, richer reports, migration tooling, broader offline where justified	**PLANNED LATER (V1)** Planned later
Stage 3: Expansion	Multi-campus depth, advanced integrations, native apps and additional operational modules	**FUTURE / VALIDATION-DEPENDENT** Validation-dependent
Stage 4: Intelligent features	AI assistance, advanced analytics and automation	**FUTURE / VALIDATION-DEPENDENT** Only after core workflows and data quality are proven

27.1 Stage gates (proposed)

Transition	Gate (qualitative)
Stage 1 to Stage 2	Pilot schools run core workflows in real use; core data quality acceptable; migration and support learnings captured; V1 priorities set from evidence; pricing hypotheses tested.
Stage 2 to Stage 3	Sustained real usage; validated demand for multi-campus, advanced integrations or native apps; support model proven.
Stage 3 to Stage 4	Core workflows and data quality proven; specific AI use cases validated; legal and privacy review complete.

28. Risks, Assumptions & Unknowns

28.1 Risk register

Risk	Why it matters	Mitigation (proposed)
Target customer not validated	Building for the wrong segment wastes effort.	Prospective-school conversations before scaling investment (Section 28.4).
Crowded, commoditized market	Feature parity does not win customers.	Focus on execution quality and switching ease; avoid feature-count competition.
Scope creep	An oversized first build delays learning.	Exclusion list and change-control rule.
RLS or tenant-isolation error	Exposure of children's data is severe.	Deny-by-default policies; automated tests; security review.
Compliance gaps	Architecture alone does not establish compliance.	Dedicated legal / product workstream (Section 23.2).
Offline complexity	Sync errors can lose or corrupt attendance.	Narrow POC; visible sync status; decision gate.
Migration friction	A poor first import damages trust.	Templates, validation, preview and reconciliation.
Report-card variety	Formats vary and can block adoption.	Bounded configurability; validate formats with pilot schools.
Support burden for a small team	Slow support drives churn.	Narrow scope; documentation; support process designed early.
Dependence on managed platforms	Pricing or policy changes affect the product.	Standard PostgreSQL; portable migrations; review of plan limits.
Unproven willingness to pay	The commercial model may not work.	Explicit pricing validation.
AI-related data risk	Sending student data to external services raises privacy issues.	Defer AI; legal review; human review of outputs.

28.2 Assumptions register

Assumption	Status	How to test
Small-to-medium schools (working range roughly 150-500 students) are a viable starting segment	[ZENCRO HYPOTHESIS]	Interviews and pilot selection.
Workflow execution matters to buyers more than additional features	[ZENCRO HYPOTHESIS]	Evaluation-criteria questions with schools.
Mobile-friendly web is sufficient for Stage 1	[ZENCRO HYPOTHESIS]	Device observation at pilot schools.
In-app announcements can replace some WhatsApp usage	[ZENCRO HYPOTHESIS]	Channel behaviour questions and pilot observation.
Basic PTM and forms provide visible value quickly	[ZENCRO HYPOTHESIS]	Pilot feedback on usage.
Attendance offline capture is technically feasible on teachers' devices	[ZENCRO HYPOTHESIS]	Proof of concept.
The chosen stack suits the team	[ZENCRO HYPOTHESIS]	Skills review and early build experience.

28.3 Unknowns register

Topic	Current position
Target customer	Hypothesis; not validated through direct interviews
Willingness to pay	Unknown
Exact ROI / savings	Unknown
Market adoption rate	Unknown
Offline breadth	Requires technical proof of concept
Native apps	Not justified for Stage 1
AI value	Potential but not sufficiently validated for MVP
Integrations	Demand varies by school; prioritise after customer validation
Compliance	Requires ongoing legal / product review; architecture alone does not establish compliance
Implementation duration	Depends on school data, migration quality and scope
Support / SLA demand	Needs commercial validation
Competitor pricing (where not public)	NOT PUBLICLY DISCLOSED - QUOTE REQUIRED
Competitor technology stacks	Not inferred; unknown

28.4 Validation plan (planned, not conducted)

> **Status**
> No interviews, surveys or calls have been conducted. The following is a plan for how the hypotheses in this document could be tested. Results must be recorded as customer evidence when they exist and must not be anticipated here.

Activity	Participants	Decision it informs
Discovery conversations	Principals, administrators, accountants and teachers at small-to-medium schools	Target segment; problem priorities; current tool use.
Workflow walk-throughs	The same roles, observing real attendance, fees and report-card work	Detailed requirements; where parallel manual processes arise.
Parent-side conversations	Parents at pilot schools	Channel preferences; parent UX requirements.
Pricing conversations	Decision-makers	Pricing basis; willingness to pay; support expectations.
Migration data review	Schools willing to share sample source data	Import design and effort.
Pilot observation	Pilot schools	Usability; support demand; V1 priorities.

29. Final Conclusion

The research does not establish that ZenCro has found a completely unserved ERP category. The Indian K-12 school ERP market is crowded and the core capability set is broadly offered. What the public evidence does support is that friction persists in the execution of ordinary school workflows, in the coordination between systems, spreadsheets, messaging and paper, and in the effort of moving between products.

For that reason, the recommended first build is not a giant ERP. It is a focused K-12 operational core: school setup, student records, attendance, fees, exams and results, communication, parent access, PTM, basic forms and import / export. The technical architecture is equally focused: Next.js, React, TypeScript, Tailwind CSS and shadcn/ui; Next.js server-side routes and actions; Supabase PostgreSQL, Auth, RLS and Storage; selected IndexedDB and Dexie offline support; Vercel; Git and GitHub; and VS Code with Antigravity.

Later capabilities, including richer integrations, admissions, learning workflows, broader offline support, advanced reporting and AI, should be added only when actual product usage and customer evidence justify the added complexity.

Priority next steps

Priority	Action	Section
1	Validate the target segment and problem priorities with prospective schools.	12, 28.4
2	Agree the Stage 1 scope baseline and the change-control rule.	14, 16
3	Run the attendance offline proof of concept as a gated experiment.	21
4	Establish the DPDP legal and product workstream.	23.2
5	Design migration and import as a first-class feature using realistic sample data.	24.3
6	Test pricing hypotheses with prospective schools.	26

30. Research Sources & Reference Categories

The underlying research used the source families below. Exact URLs and source-level citations from the earlier research are retained in the project source archive. This document does not reproduce them and does not fabricate references. Each product-level claim added to any future version of this document should be traceable to an archived source.

Source family	Use in this document	Evidence class
Ministry of Education / UDISE+	Official Indian school education statistics, including 2024-25 data (Section 4).	[REGULATORY / OFFICIAL]
Official vendor product pages and technical documentation (Indian K-12 products)	Capability patterns; positioning; offline, AI and multi-campus claims (Sections 5, 6).	[VENDOR CLAIM]
Official technical documentation for open-source and international school-management products	Workflow benchmarking and implementation patterns (Section 7).	[VENDOR CLAIM] / [ANALYST OBSERVATION]
Independent software-review platforms and public customer reviews	Problem and evidence mining (Section 8).	[INDEPENDENT EVIDENCE] / [CUSTOMER EVIDENCE]
School / institutional documents, procurement material, educational-organisation evidence	Workflow requirements; buyer and procurement patterns (Sections 8, 9).	[INSTITUTIONAL EVIDENCE]
Public discussion forums and app-store evidence	Customer-experience themes, used cautiously (Section 8).	[CUSTOMER EVIDENCE]
Public technical documentation	Benchmarking architecture and implementation patterns (Sections 19-21).	[ANALYST OBSERVATION]

Source-use rule

Vendor claims are treated as vendor claims. Customer evidence is treated as customer evidence. Unknowns are not converted into facts. Public research is used to form product hypotheses, not to claim statistical market certainty.

31. Appendices

Appendix A. Competitor verification worksheet (template)

Use this template to record source-level evidence for each product. Every cell starts as UNKNOWN. Change it to YES only when a source is recorded; use NO only when a reliable source states the capability is absent. Public pricing that cannot be found stays as: NOT PUBLICLY DISCLOSED - QUOTE REQUIRED.

Field	Value to record	Default
Product name	As shown by the vendor	n/a
Segment served	As stated by the vendor	UNKNOWN
SIS / records	YES / NO / UNKNOWN + source	UNKNOWN
Attendance (incl. offline claim)	YES / NO / UNKNOWN + source	UNKNOWN
Fees / payments	YES / NO / UNKNOWN + source	UNKNOWN
Exams and report cards; customisation	YES / NO / UNKNOWN + source	UNKNOWN
Parent / teacher apps	YES / NO / UNKNOWN + source	UNKNOWN
Communication and targeting	YES / NO / UNKNOWN + source	UNKNOWN
Digital forms / consent; PTM booking	YES / NO / UNKNOWN + source	UNKNOWN
Multi-campus	YES / NO / UNKNOWN + source	UNKNOWN
Integrations (payments, accounting, messaging)	YES / NO / UNKNOWN + source	UNKNOWN
Migration / export terms	As stated by the vendor	UNKNOWN
AI / automation claims	As stated (label VENDOR CLAIM)	UNKNOWN
Public pricing	As published	NOT PUBLICLY DISCLOSED - QUOTE REQUIRED
Technology stack	Only if stated in vendor documentation	UNKNOWN (never inferred)

Appendix B. Decision log (baseline decisions carried into this document)

Decision	Basis
Scope is K-12 school management only; college / university ERP excluded	Project scope decision
No claim of an unserved category or of no competitors	Research conclusion
Positioning based on execution quality, not feature count	Research conclusion
Stage 1 limited to the operating loop in Section 14	Product strategy
Modular monolith on Next.js with Supabase; no separate backend, microservices, Kubernetes, Redis, queue or vector database in Stage 1	Architecture direction
Offline limited to attendance, gated by a POC	Architecture direction
AI deferred; never the source of truth for core records	Product and architecture principle
DPDP compliance treated as a legal / product / process workstream	Security and privacy position
No competitor pricing or technology stack estimated	Research discipline